

Mathematical Foundations of Political Methodology

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Interpreting Causal Effects

Rubin Causal Model:

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- $Y_i(1)$: response under treatment

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$$ATE = E[Y(1) - Y(0)]$$

Observational Studies and Causal Inference

Experimental studies:

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- Treatment under control of analyst
- Random assignment, estimate

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The Problem of Selection

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Experiments:

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Observational studies:

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- **Problem:** how do we learn about counterfactuals in the face of selection?

Manski Bound

What can we learn in the face of selection?

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- Interval that contains true ATE with probability 1
- Reduces interval length to 1
- Requires no additional assumptions

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Note that:

$$0 \leq E[Y(1)|T = 0] \leq 1$$

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and has length 1.

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- $\pi = 0.06$
- $E[Y(1)|T = 1] = 18/21 = 0.857$
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- Further assumptions can narrow bounds
- **Law of Decreasing Credibility:** the credibility of inference decreases with the strength of the assumptions maintained

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 - 2) $E[Y(0)|D = 1, S] = E[Y(0)|D = 0, S]$

Proposition

Suppose there exists S such that 1) and 2) hold. Then, we can obtain unbiased estimates for

- 1) $E[ATE|S = 1] \equiv ATE$, Given $S = 1$
- 2) $E[ATE|S = 0] \equiv ATE$, Given $S = 0$
- 3) ATE

Idea:

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- 1) Average Treatment effect with strata, $S = 1$

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 - Calculate Average(Treatment) - Average(Control) within each strata,
 $E[ATE|S]$

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- 3) Average Treatment Effect
 - Calculate Average(Treatment) - Average(Control) within each strata, $E[ATE|S]$
 - $ATE = \sum_{s=0}^1 E[ATE|S] \times Pr(S)$

Proof

$$E[Y(1)|S = 1, D = 1] - E[Y(0)|S = 1, D = 0]$$

Proof

$$\begin{aligned} & E[Y(1)|S = 1, D = 1] - E[Y(0)|S = 1, D = 0] \\ &= \{ \Pr(D = 1|S = 1)E[Y(1)|S = 1, D = 1] \\ &\quad + \Pr(D = 0|S = 1)E[Y(1)|S = 1, D = 1] \} \\ &\quad - \{ \Pr(D = 1|S = 1)E[Y(0)|S = 1, D = 0] \\ &\quad + \Pr(D = 0|S = 1)E[Y(0)|S = 1, D = 0] \} \end{aligned}$$

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Causal Inference via Stratification: Example

We are interested in the causal effect of **incumbency** on reelection.

- $T = 1$, Incumbent
- $T = 0$, Challenger

$Y_i(T)$ result of election.

Incumbency obviously not assigned at random.

But suppose we have a dichotomous measure of candidate quality

- $S = 1$, High quality
- $S = 0$, Low quality

And that incumbency is as good as random, given S .

We're interested in obtaining

$$ATE = E[Y(1) - Y(0)]$$

What if we don't condition on candidate quality?

Joint Distribution of Treat and Strata

	$T = 1$	$T = 0$	$P(S)$
$S = 1$	0.4	0.2	0.6
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 E[Y(1)|T = 1] - E[Y(0)|T = 0] &= 0.64 - 0.38 \\
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Why?: those with bigger treatment effects more likely to select into treatment

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Confound effect of T with differences across S

With Stratification

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$$\sum_{i=1}^2 (E[Y(1)|S = i, T = 1] - E[Y(0)|S = i, T = 0]) \Pr(S = i) =$$

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Stratification and Matching

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Exact Matching, Basic idea:

- Identify all **characteristics** (covariates) that affect both outcome and treatment assignment
- For all treated units, identify **control** unit with same characteristics
- Exact match: units in the same strata

What do we do?

Most strata are empty, or only a few observations

- Bias-Variance tradeoff
 - Bias: assume same casual effect across strata
 - Variance: assume different causal effect across strata
- Modeling $E[Y|X_1, X_2, X_3, \dots, X_K]$
 - Nonparametric (loess): different curse of dimensionality problem
 - High dimensional space is sparse, hard to borrow across

Solution: specify a model of how covariates relate to treatment

Linear Regression

Consider one continuous covariate S_i and a continuous dependent variable Y_i .

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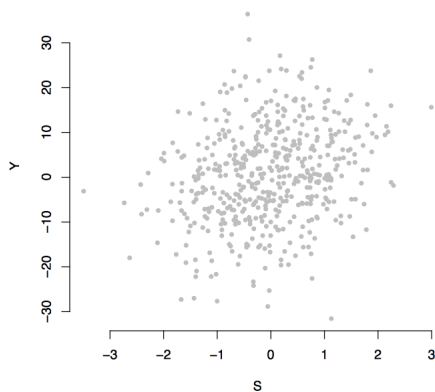
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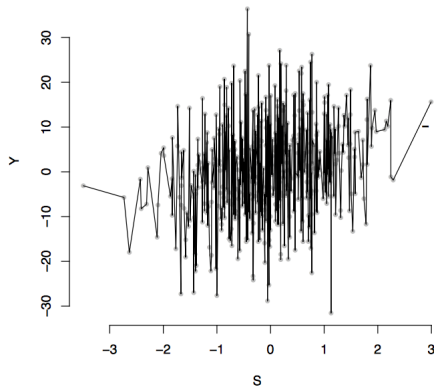
What does this say?

- **Stratifying** (conditioning): examining means of Y given values of S
- **Borrowing** information across bins:
 - **Assuming** that means have a **global** and **linear** movement with S , β_1

Linear Regression

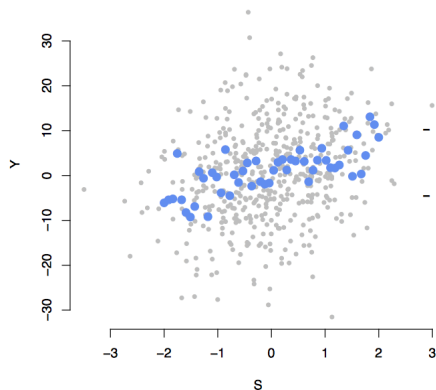


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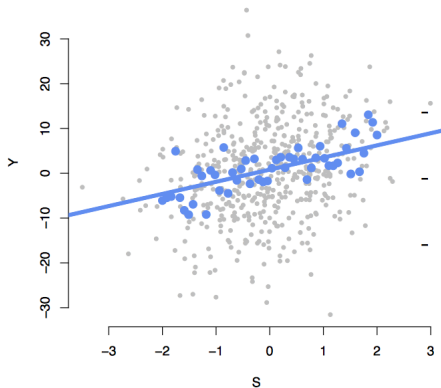
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(0.1)

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We are going to find the β_0^*, β_1^* that minimize the sum of squared residuals,

(0.1)

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$$(\beta_0^*, \beta_1^*) = \underset{\beta_0, \beta_1}{\operatorname{argmin}} \sum_{i=1}^N \epsilon_i^2 \tag{0.1}$$

For every random variable Y and S we can always write the random variable as,

$$Y_i = \underbrace{Y_i - E[Y|S]}_{\epsilon_i} + E[Y|S]$$

$$Y_i = E[Y|S] + \epsilon_i$$

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Where we have used our assumption about $E[Y|S] = \beta_0 + \beta_1 S_i$

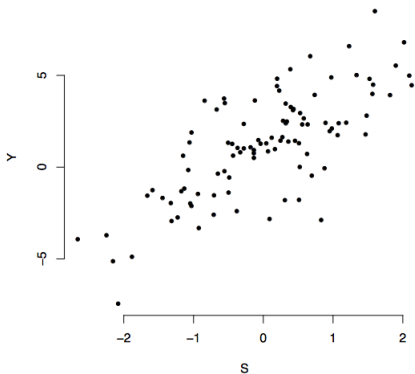
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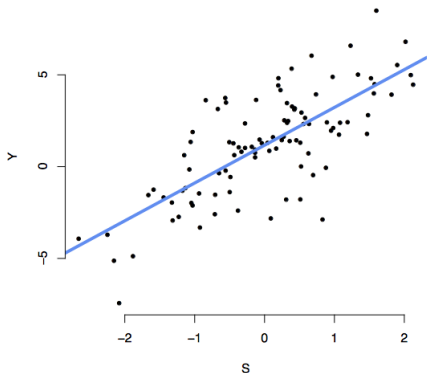
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